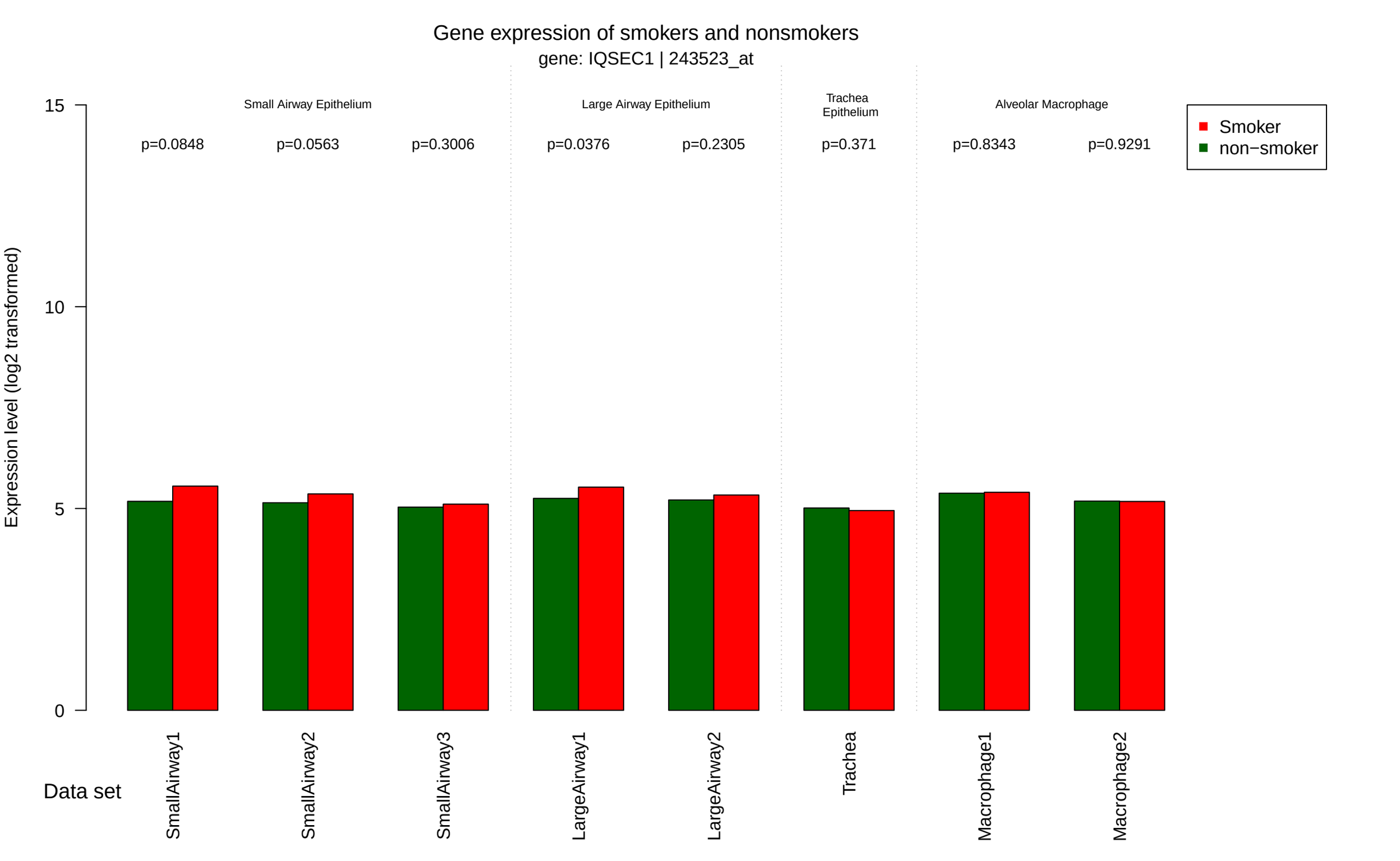
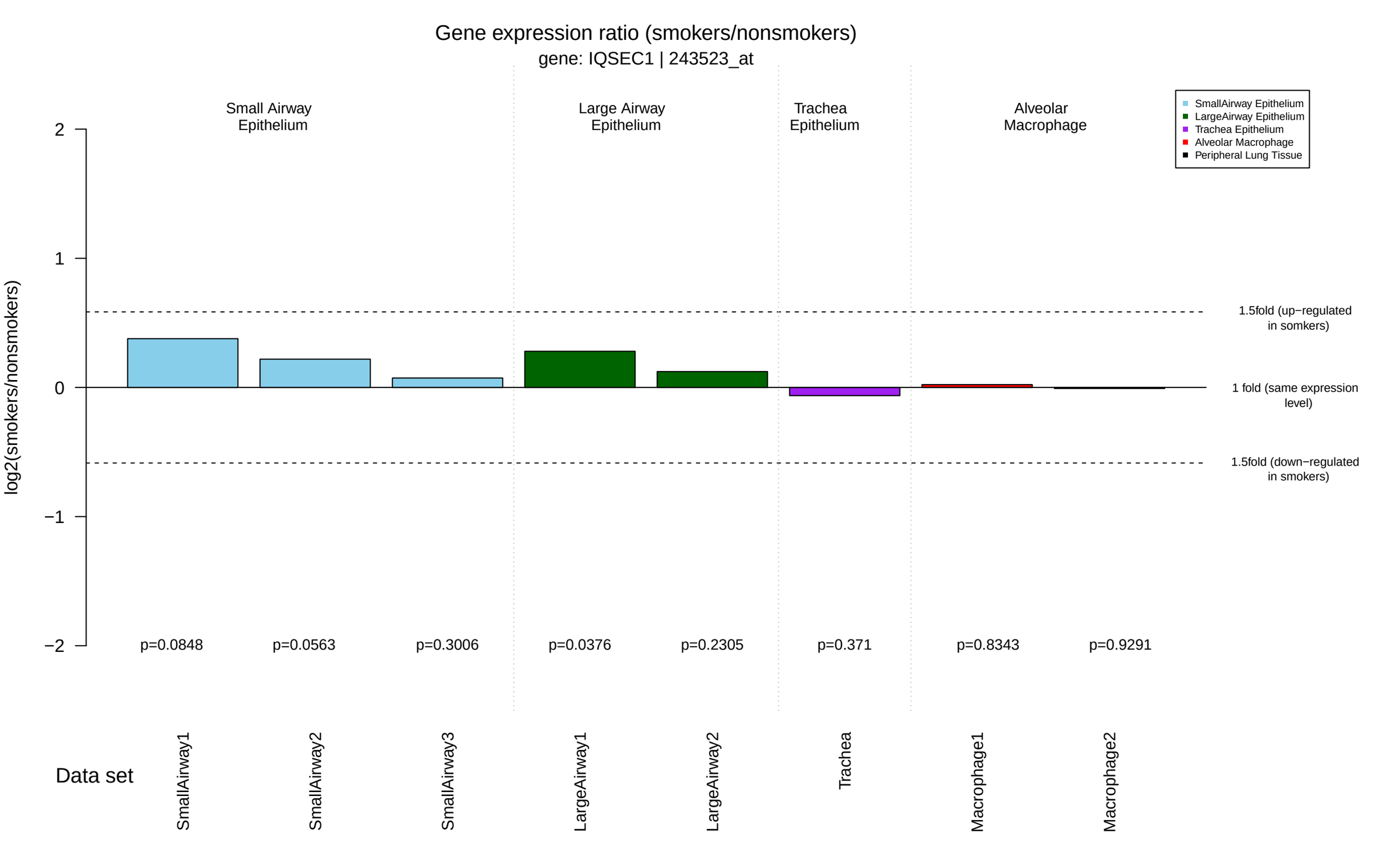
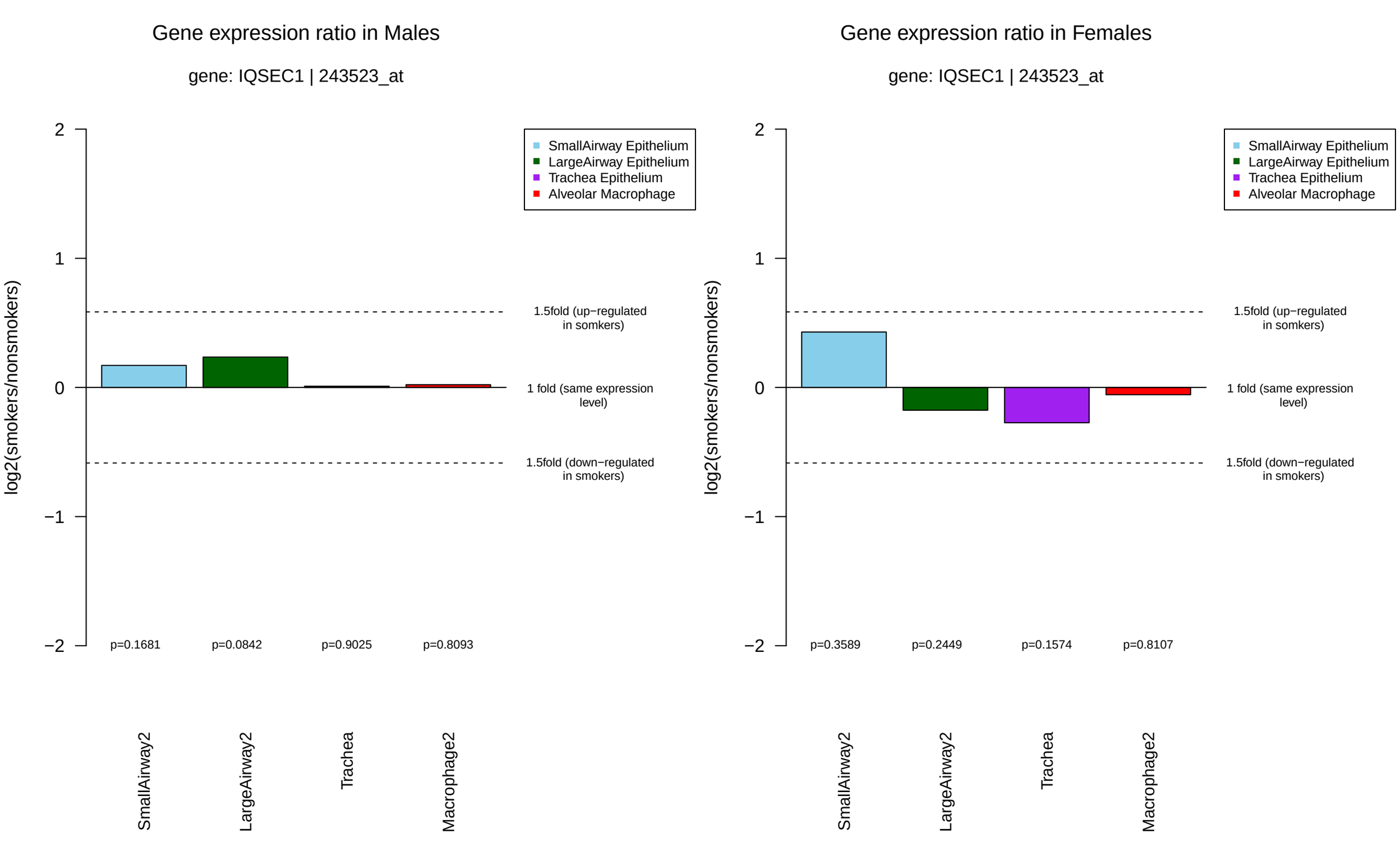
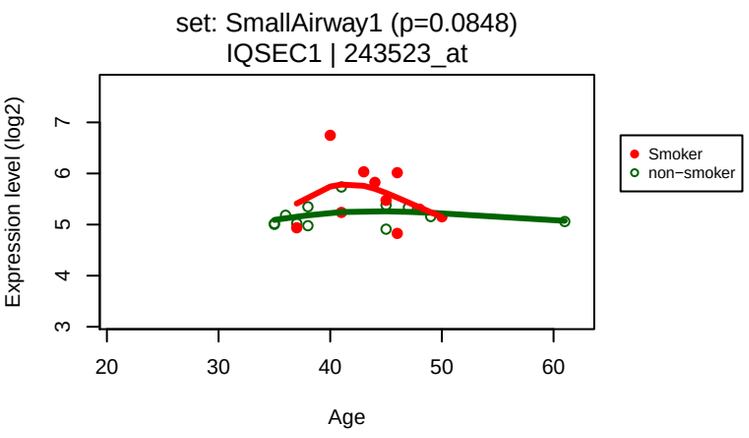
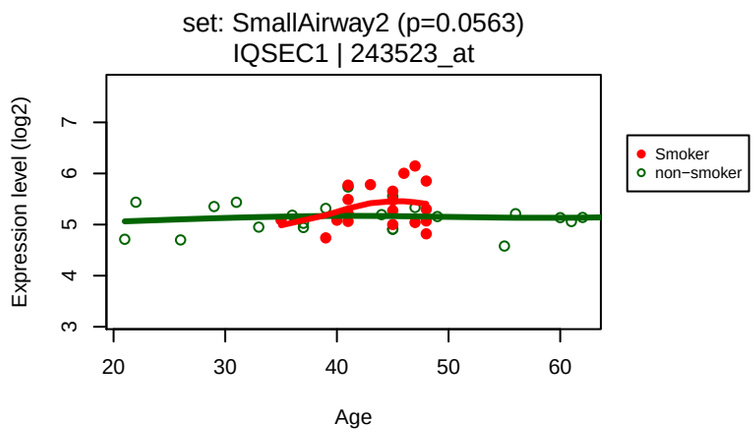
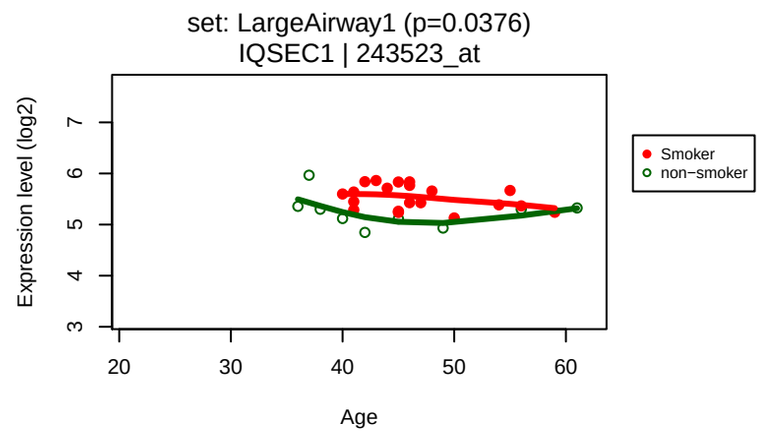

Gene expression of smokers and nonsmokers
gene: IQSEC1 | 243523_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0848
p=0.0563
p=0.3006
p=0.0376
p=0.2305
p=0.371
p=0.8343
p=0.9291
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

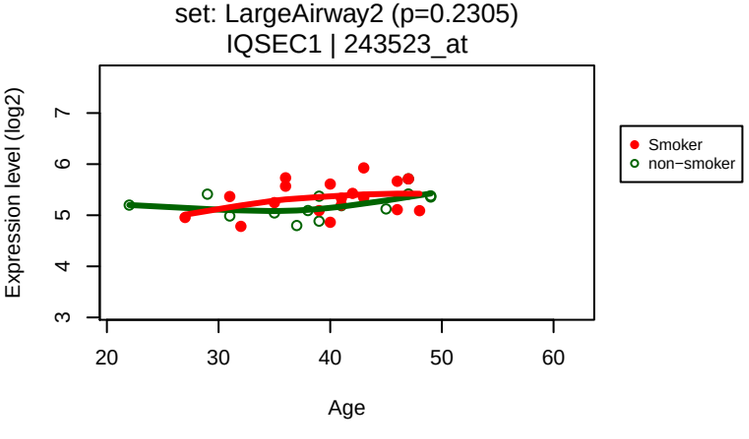

Gene expression ratio (smokers/nonsmokers)
gene: IQSEC1 | 243523_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0848
p=0.0563
p=0.3006
p=0.0376
p=0.2305
p=0.371
p=0.8343
p=0.9291
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

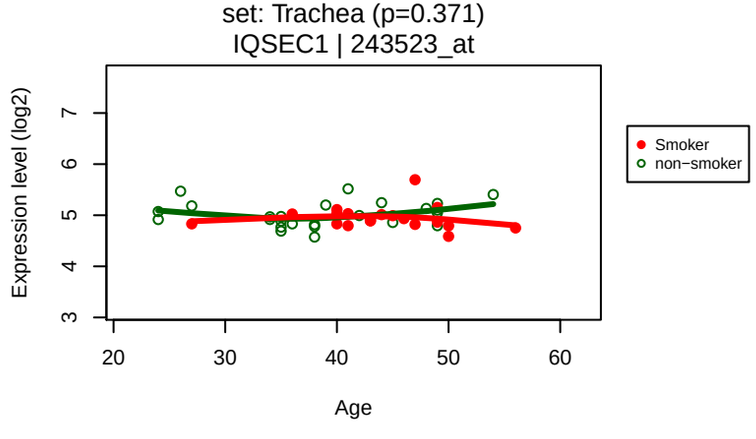

Gene expression ratio in Males
gene: IQSEC1 | 243523_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1681
p=0.0842
p=0.9025
p=0.8093
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IQSEC1 | 243523_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3589
p=0.2449
p=0.1574
p=0.8107
SmallAirway2
LargeAirway2
Trachea
Macrophage2

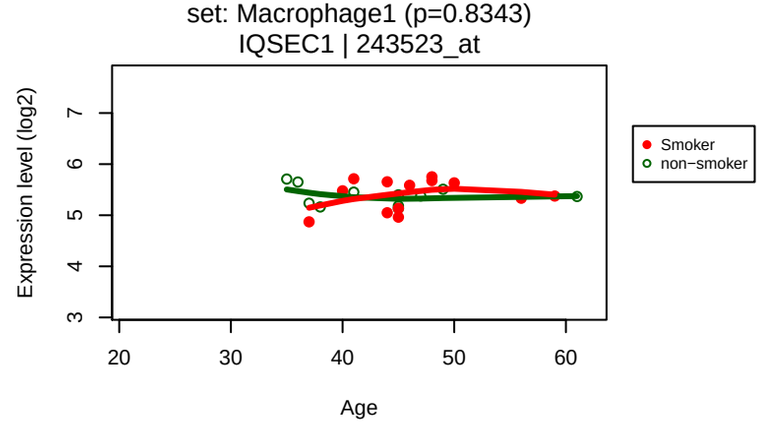

set: SmallAirway1 (p=0.0848)
IQSEC1 | 243523_at
Expression level (log2)
Age
Smoker
non−smoker

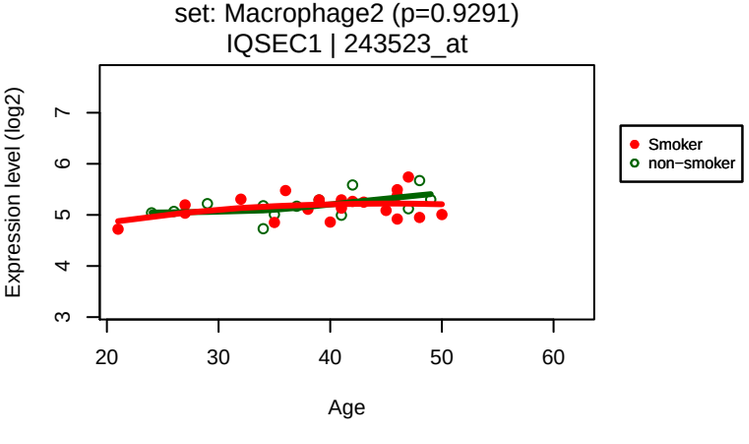

set: SmallAirway2 (p=0.0563)
IQSEC1 | 243523_at
Expression level (log2)
Age
Smoker
non−smoker

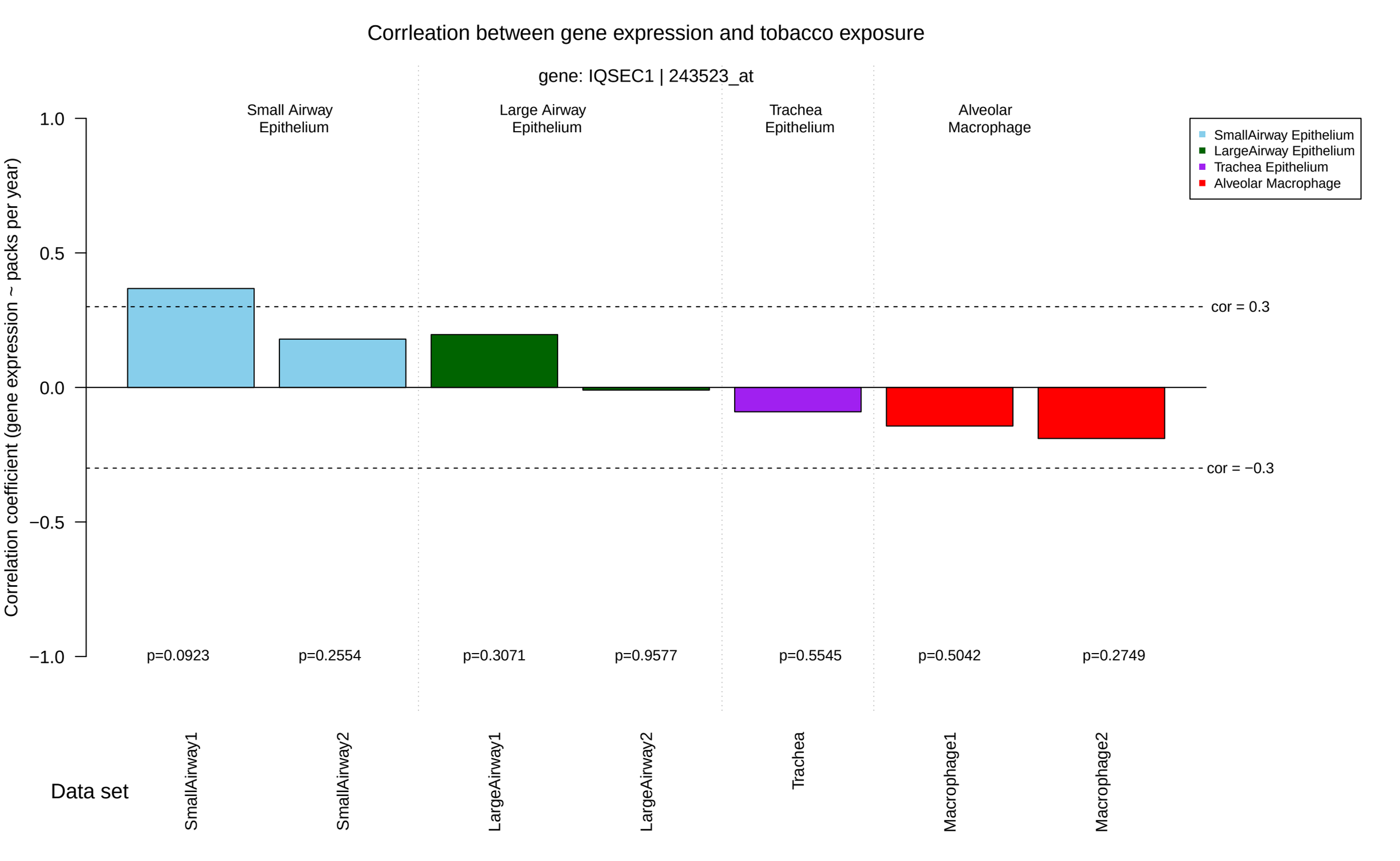

set: LargeAirway1 (p=0.0376)
IQSEC1 | 243523_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2305)
IQSEC1 | 243523_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.371)
IQSEC1 | 243523_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8343)
IQSEC1 | 243523_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9291)
IQSEC1 | 243523_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IQSEC1 | 243523_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0923
p=0.2554
p=0.3071
p=0.9577
p=0.5545
p=0.5042
p=0.2749
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2